SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: ITA 0607 **COURSE NAME: MACHINE LEARNING**

COURSE OUTCOME COVERED

CO1: *Learning Problems, Perspectives and Issues, Concept Learning, Version*

Spaces & Candidate Elimination, Inductive Bias, Decision Tree Learning, Representation & Heuristic Search (BL1, BL2)

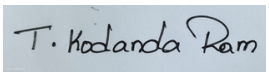
Assessment Tool Weightage: 20%

QUESTIONS: 15 **Total Marks: 25**

Annexure A

CONCEPT MAPPING

Code of Conduct:
I T . KODANDA RAM (192524162) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Annexure A

Short Answer Test

1. With the necessary sketch, explain all the possible blocks in the machine-learning system. Give example for showing the concept of the supervised and Un-supervised learning methods in the machine learning scheme. (5 Marks)

11. Explain the architecture of a typical machine learning pipeline with a neat diagram. Discuss the role of each stage including data collection, preprocessing, feature engineering, model training, evaluation, and deployment. Also, differentiate between training error and generalization error with suitable examples. (5 Marks)

12. Design a machine learning model for spam email classification using labeled data
(a) Explain the choice of algorithm (e.g., Naïve Bayes, SVM, or Logistic Regression).
(b) Describe the preprocessing steps such as tokenization, stop-word removal, and vectorization.
(c) Discuss evaluation metrics such as accuracy, precision, recall, and F1-score. (5 Marks)

14. Develop a machine learning system for predicting house prices using features like location, size, number of rooms, and amenities. (5 Marks)

(a) Justify the choice of regression model (Linear Regression, Decision Tree, or Random Forest).
(b) Explain feature scaling and handling missing data.
(c) Describe how overfitting can be detected and avoided.

15. Explain the concept of clustering in unsupervised learning. (5 Marks)

(a) Describe the working of the K-Means algorithm with a step-by-step example.
(b) Explain how the value of *K* is chosen using the Elbow Method.
(c) Compare K-Means with Hierarchical Clustering.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	**30**	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	**20**	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	**20**	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	**15**	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	**15**	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

CO1 - AT2 -
Concept Mapping

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Course : ITA0607 - Machine Learning

Machine Learning System

Main Block / Components

- **Data Collection** → - Read world data
  - Labled / unlabled
- **Data Preprocessing** → - clean & Transform data
  - Handle missing values
- **Feature selection / Feature Engineering** → - select important features
  - Convert to useful format
- **Learning Algorithm** → - supervised / unsupervised
  - Train a model
- **Model Evaluation** → - Test performance
  - Accuracy / Error
- **Prediction / output** → - Final Model
  - Make predictions

Types of Learning Methods

Supervised Learning

- uses labled data.
- Learns input → output mapping
- Used for classification and regression.

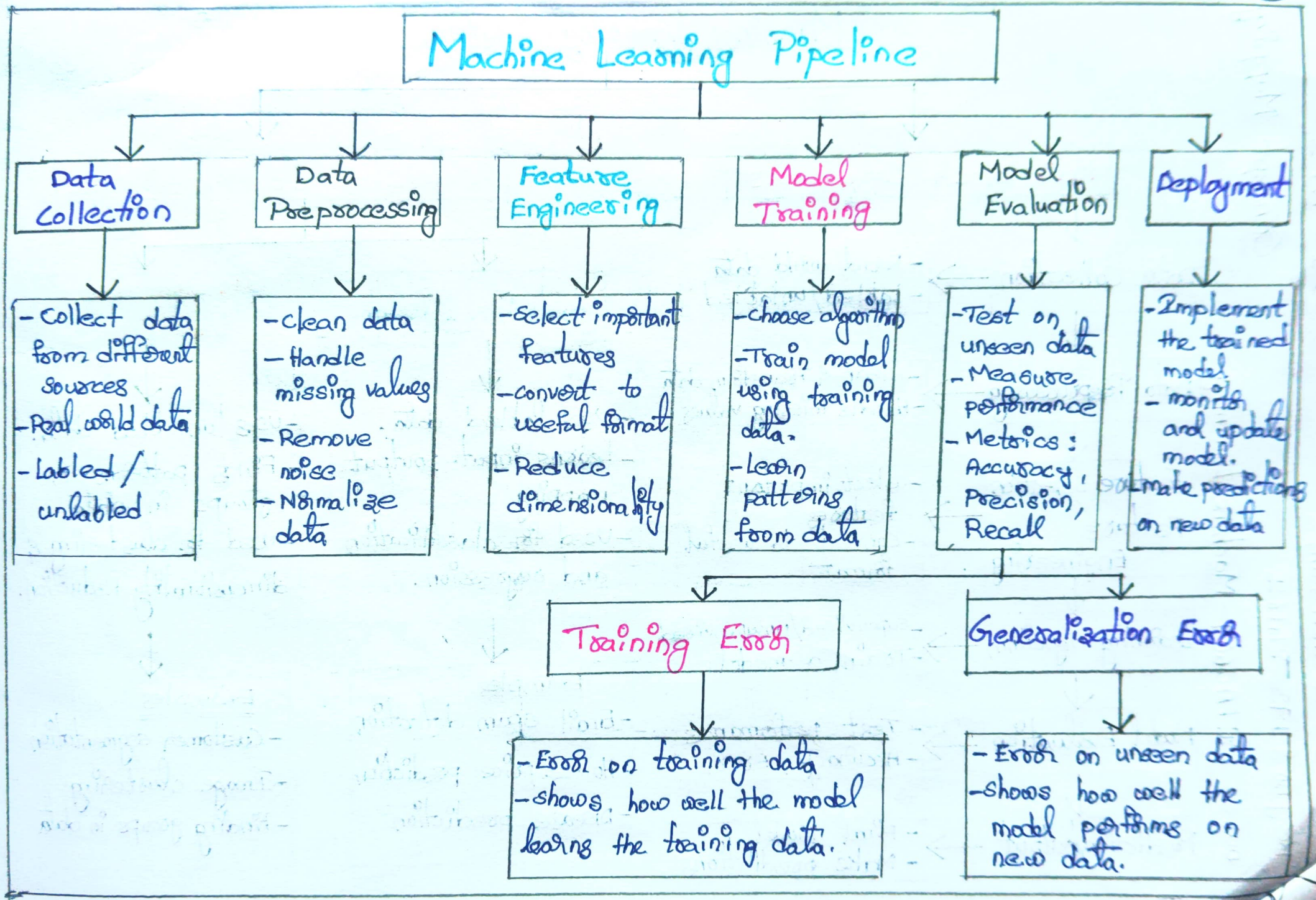
Examples
- Email spam detection
- House price prediction
- Disease prediction

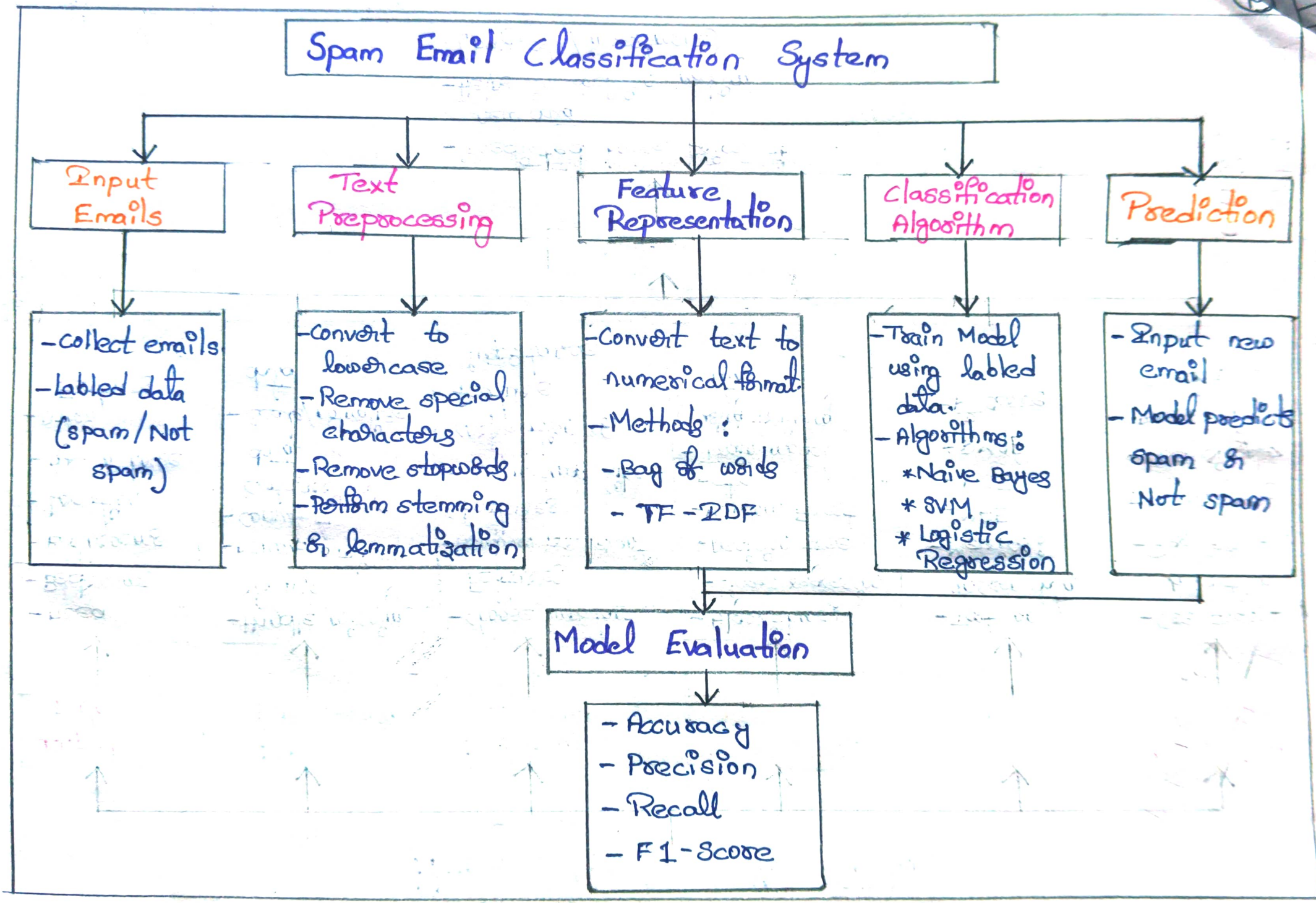
unsupervised Learning

- uses unlabled data.
- Finds patterns & groups in data
- Used for clustering & dimensionality Reduction.

Examples
- Customer segmentation
- Image clustering
- Finding groups in data

Machine Learning Pipeline




Spam Email Classification System
Input Emails
Text Preprocessing
Feature Representation
Classification Algorithm
Prediction
-collect emails
-Labled data (spam/Not spam)
-Convert to lowercase
-Remove special characters
-Remove stopwords
-Perform stemming & lemmatization
-Convert text to numerical format
-Methods :
-Bag of words
- TF-IDF
-Train Model using labled data.
-Algorithms:
*Naive Bayes
* SVM
* Logistic Regression
- Input new email
- Model predicts spam & Not spam
Model Evaluation
- Accuracy
- Precision
- Recall
- F1-Score

House Price Prediction System

Input Data
- Area
- Bedrooms
- Bathrooms
- Location
- Age of house
- other Features

Data Preprocessing
- Handle missing values
- Remove outliers
- convert categorical data.
- Scale/normalize data

Feature Selection
- Choose important Features
- Remove irrelevant features
- Use methods like correlation & feature importance

Regression Model
- Algorithms:
  - Linear regression
  - decision tree
  - Random Forest
  - etc
- Train model using training data

Model Evaluation
- Test on unseen data
- Metrics:
  - MAE
  - MSE
  - RMSE
  - R^2 score

Overfitting Prevention
- Use cross-validation
- Regularization
- Prune trees
- use simpler model if needed

Output
- Predicted house price for new data
- Helps in decision making (buy/sell/investment)

Clustering (Unsupervised Learning)

What is clustering?

- Groups similar data points
- No labels

K-Means Clustering

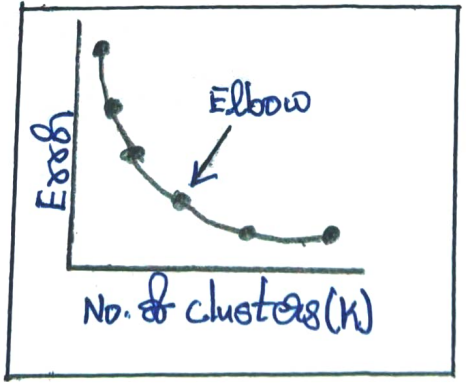
- Divides data into K clusters.
- Each cluster has a center (centroid)

steps in K-Means

1) choose K
2) Initialize centroids
3) Assign points to nearest centroid
4) Update centroids
5) Repeat until no change.

Elbow Method

- Run K-Means for different K values
- plot error (within-cluster sum of squares)
- choose K at the elbow point



K-Means vs Hierarchial Clustering

K-Means	Hierarchial
- Flat clustering	- Builds a tree (dendrogram)
- Need to specify K	- No need to specify K
- Faster	- slower
- Good for large data	- Good for small data